THE CENTER OF GRAVITY

CONOR DYSON

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PhD candidate at the Centre of Gravity within the Niels Bohr Institute, University of Copenhagen. Supervised by Maarten van de Meent and Vitor Cardoso. Research focuses on post-adiabatic gravitational self-force methods for modelling extreme and intermediate mass ratio inspirals, with applications to gravitational-wave astronomy and black hole environments.

CURRENT POSITION

PhD Fellow, Centre of Gravity, Niels Bohr Institute, University of Copenhagen

RESEARCH INTERESTS

Black Holes, Gravitational Waves, General Relativity, Extreme Mass Ratio Inspirals, Environments, Self-Force, Dark Matter, and Relativistic Fluids

EDUCATION

PhD	**Niels Bohr Institute, University of Copenhagen**, *Post-Linear Effects in Extreme Mass Ratio Inspirals*, Supervisors: Maarten van de Meent *&* Vitor Cardoso	2022–2026
	Massachusetts Institute of Technology, Visiting Student (1 semester), Host: Scott Hughes	2025
MASt	**DAMTP, University of Cambridge**, Applied Mathematics. *Perturbations of a Kerr Black Hole*, Essay supervisor: Harvey Reall	2021–2022
BSc	**University College Dublin**, Applied and Computational Mathematics. *Ray Tracing and Neutron Star Visualisation*, Supervisor: Sarp Akcay	2017–2021
	University of Edinburgh, Visiting Student (1 semester)	2020

AWARDS & HONOURS

Best Scientific Poster, Black Holes Inside and Out, Strong gravity group.	2024
Ad Astra Scholarship, University College Dublin	2018-2021
BSc Stage 4 Scholarship, University College Dublin	2020
Fr. Ciaran Ryan Prize, University College Dublin	2020

PUBLICATION SUMMARY

5 publications (incl. 1 PRL), 4 as first or corresponding author, 117 citations (h-index: 5)

INVITED SEMINARS

Environments Discussion Panel, LISA Waveform Working Group Meeting, **SISSA, Trieste** (expected)	Feb. 2026
Environments and Fundamental Physics Discussion Panel, Capra Meeting, **University of Southampton**	July 2025
Self-Force Seminar, **University of Southampton**	July 2025
Coffee Talk, **Institute for Advanced Study**	May 2025
Coffee Talk, **Centre for Theoretical Physics, MIT**	May 2025
ACR group meeting presentation, **Albert Einstein Institute**	Nov. 2023

CONTRIBUTED TALKS

24th International Conference on General Relativity, **Glasgow, Scotland**	July. 2025
28th Capra Meeting on Radiation Reaction in GR, **Southampton, England**	July. 2025
YITP long-term workshop in Gravity and Cosmology, **Kyoto, Japan**	Feb. 2024
26th Capra Meeting on Radiation Reaction in GR, **Copenhagen, Denmark**	June. 2024

POSTER PRESENTATIONS

Lisa Symposium, **Dublin, Ireland**	July. 2025
Black Holes Inside and Out, **Copenhagen, Denmark**	August. 2025

ORGANISATION

Strong Group Weekly Meetings and Seminars, Copenhagen (Denmark)	2024–Present
Strong Group Retreat, Fuglsang Manor (Denmark)	June 2025
Niels Bohr International Academy Barbecue, Copenhagen (Denmark)	Sept. 2024
Post-linear Meetings, Niels Bohr Institute, Copenhagen (Denmark)	2024
Capra Conference on Radiation Reaction in General Relativity, Copenhagen (Denmark)	June 2023

INTERNATIONAL VISITS

Massachusetts Institute of Technology (3 months). *Host: Scott Hughes*	2025
Johns Hopkins University (1 week). *Host: Daniel D'Orazio*	2025
University of Southampton (1 week). *Host: Adam Pound*	2025
Institute for Advanced Study (2 days). *Host: Giovanni Tomasellii*	2025
Yukawa Institute for Theoretical Physics, Kyoto University (5 weeks)	2024
Albert Einstein Institute (1 week). *Host: Maarten van de Meent*	2023

COLLABORATIONS

Multi-Scale Self-Force Collaboration	2023–Present
LISA Consortium, Project Coordinator & Core Member	2025–Present
Black Hole Perturbation Toolkit, Contributor	2023–Present

TEACHING

Master student supervision: Manon Johansson (15 ECTS)	2025/26
Master student supervision: Ziying Ma (7.5 ECTS)	2025/26
TA: Stochastic Inverse Problems (Lecturer: K. Mosegaard)	2023/24
TA: Computational Science (Lecturer: J. Samsing)	2023

(Teaching at the University of Copenhagen)

SOFTWARE

Expert in Mathematica. Proficient in Python and MATLAB. Proficient with xAct and the Black Hole Perturbation Toolkit

SERVICE

LISA Consortium: Project Coordinator
Referee for **PRD**, **PRL**, **CQG**, and **SciPost**